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EV Charging Scheduling Agent: Agentic AI for Smart EV Charging Scheduling

Peter Quawas & Ryan Luo

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Motivation

- Charging systems must balance competing objectives: Minimizing Time-of-Use (TOU) energy costs, meeting user demand, and respecting charger capacity limits
- Variable TOU pricing and rolling time horizons mean naive scheduling leads to unmet demand, grid overloads, or unnecessary costs
- We benchmark three approaches: (1) a convex optimizer, (2) an LLM only optimizer, and (3) an agentic LLM + optimizer pipeline

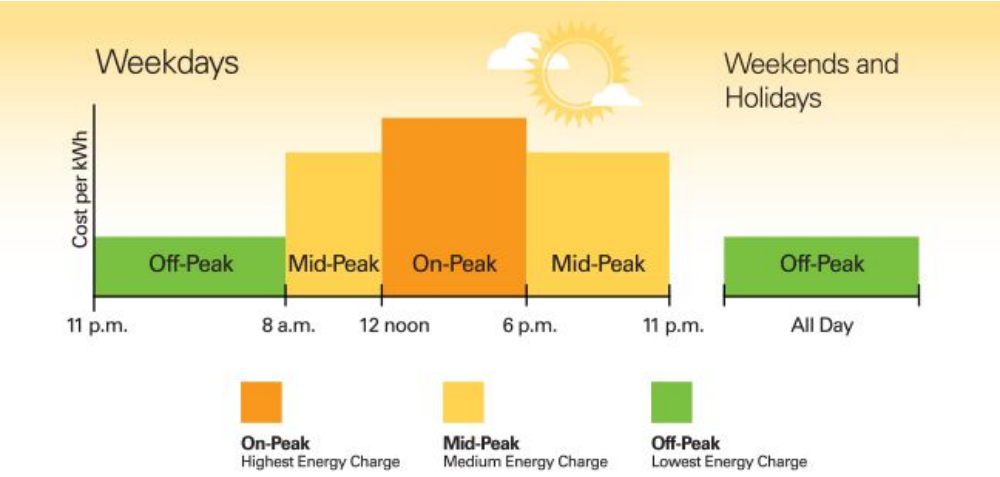

Figure 1: Time of Use (TOU) pricing

Introduction

Inputs & Outputs

Inputs:

- EV sessions: arrival/departure, requested energy, max charge rate
- Site config: Max Power = 50kW over 96 time steps (15m intervals)
- TOU rates: off peak $0.12/kWh, peak $0.45/kWh
- LLM & Agent: Problem parameters delivered as a structured or natural language request

Outputs:

- Power schedule (p[i, t], kW per EV session per time step)
- Metrics: total cost, peak load, unmet energy, % sessions fully served, # of constraint violations
- LLM/Agent: natural language summary based on these metrics

Data

- Utilized Caltech Adaptive Charging Network (ACN) Data
- Numerical solver and agent uses data parsed into inputs
- Connection/disconnect, energy, max power on a 96-step horizon (15-min resolution)

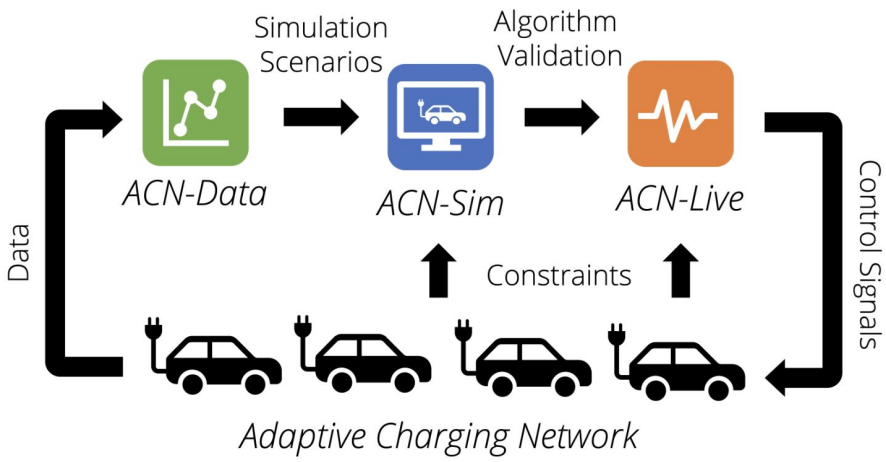

Figure 2: ACN Roadmap

Methodology (Numerical Solver)

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CVXPY

Convex optimization in CVXPY: goal to minimize TOU energy cost and minimize penalty on unmet energy over 96 time steps

Input

- Standardized sessions (arrival, departure, energy, max power per session)
- Site config: P_max, 96 times steps of 15min
- TOU rates (price per kWh per step)

Output

- Schedule (power p[i,t] in kW)
- Metrics: cost, peak load, unmet energy, % sessions fully served, constraint violations

Constraints

Availability windows, per-charger power limits, site cap, energy delivery

Constraint Checker (Used by all methods)

- Checks any schedule against:
  1. Availability: `p[i,t] = 0` for `t` outside session window
  2. Per-charger limits: `0 ≤ p[i,t] ≤ max_power_kw`
  3. Site cap: `∑_i p[i,t] ≤ P_max(t)` at each `t`
  4. Energy: no over-delivery; reports unmet energy per session
- Returns: feasible flag, list of violations, per-session unmet, peak load

Methodology (LLM Baseline)

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Architecture

- Single shot GPT-4o asked to output full 96 step schedule matrix

Data → Prompt (Parsing)

- Prompt includes sessions table (arrival and departure_idx, energy_kwh, max_power_kw), site cap, TOU rates, and constraints
- Output format enforced in prompt:
  - Session i: p0 … p95
- Parser extracts schedule matrix

Optimization problem (described to the LLM)

- **Objective**: Minimize TOU cost respecting availability, per-charger limits, and site-cap constraints
- Do output constraint checking (same as numerical solver)

Faithfulness

- Explanation should rely only on metrics (unmet energy, violations) to reveal when LLM schedules are invalid

Methodology (Agent)

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Architecture

- LLM (GPT-4o) + solve_ev_schedule tool (CVXPY optimizer)
- Agent converts NL input into structured session data before calling solver and checker (same as numerical solver)

Tool (CVXPY Solver)

- Minimizes TOU + unmet energy penalty under availability, per charger, and site constraints, returning cost, peak load, unmet demand, % of sessions served, constraint violations.

Novel aspects

- Agent grounds decision in numerical optimizer rather than directly generating text

Prompt Design

- Agent parses user input to extract essential information needed to create charging schedule
- If minimum information is not provided, agent will ask follow ups in structured manner
- When sufficient information is provided, agent infers reasonable defaults and states assumptions

Methodology (Agent) Safety Replies

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You

I have 1 EV

Agent

To compute the optimal charging schedule, I need the following information:

- EV-1: arrival time
- EV-1: departure time
- EV-1: energy needed (kWh)

Why this information is needed (campus charging context):
- Arrival time: When you park at the campus lot
- Departure time: When you're leaving campus
- Energy needed: Based on your commute distance (~10-20 kWh typical)

Please provide this information so I can compute an optimal schedule.
If some details are unavailable, let me know which ones and I'll use reasonable estimates based on typical campus charging patterns.

Example: "EV 1 arrives at 9am, leaves at 5pm, needs 15 kWh."
Or with unknowns: "EV 1 arrives in the morning, not sure when I'm leaving, need about 20 kWh for my commute."

You

I have 3 EVs. EV 1 arrives at 7pm and needs 10kWh. EV 2 arrives at 6pm and leaves at 10pm. EV 3 leaves at midnight and needs 15kWh

Agent

The charging schedule successfully serves all 3 EV sessions fully, with no unmet energy requirements. The total cost of energy is $6.18, and the peak load during the schedule is 19.2 kW. This means that all sessions received the energy they needed within their respective time windows, and the schedule adhered to the site power cap of 50 kW. (Note: I inferred some values based on context: EV-1: 7pm arrival, needs 10kWh → departure 10pm (10kWh/7kW = ~1.43h, rounded to 3h stay); EV-2: 6pm arrival, 10pm departure → 4h stay, max deliverable = 4h * 7kW = 28kWh, inferred 10kWh as typical commute; EV-3: Departure at midnight, needs 15kWh → assumed 9pm arrival (15kWh/7kW = ~2.14h, rounded to 4h stay))

Inferred values:
EV-1: 7pm arrival, needs 10kWh. Max deliverable in 3h (7pm-10pm) is 21kWh, so departure set to 10pm.
EV-2: 6pm arrival, 10pm departure. 4h stay allows for 28kWh max. Typical evening stay is 3-5h, so inferred energy is 16kWh (4h * 4kW average).
EV-3: Departure at midnight, needs 15kWh. Assuming typical 4h evening stay, arrival inferred as 8pm (15kWh/7kW = ~2.14h, rounded to 4h stay).

Cost:	**$6.18**	Peak Load:	**19.2 kW**
Sessions:	**3**	Served:	**100%**

Methodology (Agent) Successful Replies

EV Charging Schedule Agent
Powered by OpenAI + CVXPY optimizer

You

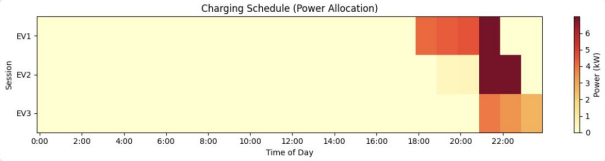
I have 3 EVs: EV1 arrives 6pm leaves 10pm needs 20 kWh, EV2 arrives 7pm leaves 11pm needs 15 kWh, EV3 arrives 8pm leaves midnight needs 10 kWh

Agent

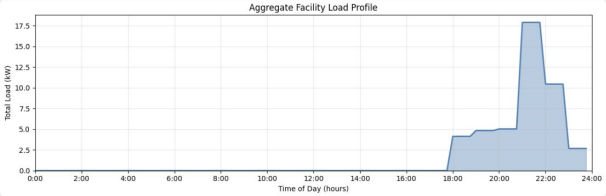
The charging schedule successfully serves all three EV sessions fully, with no unmet energy requirements. The peak load during the charging period is 17.88 kW, and the total cost of energy is $10.02. This means that all sessions received the required energy within their respective time windows, and the schedule efficiently avoided the more expensive time steps as much as possible.

Cost:	**$10.02**	Peak Load:	**17.88 kW**
Sessions:	**3**	Served:	**100%**

📊 Charging Schedule (sessions × time)

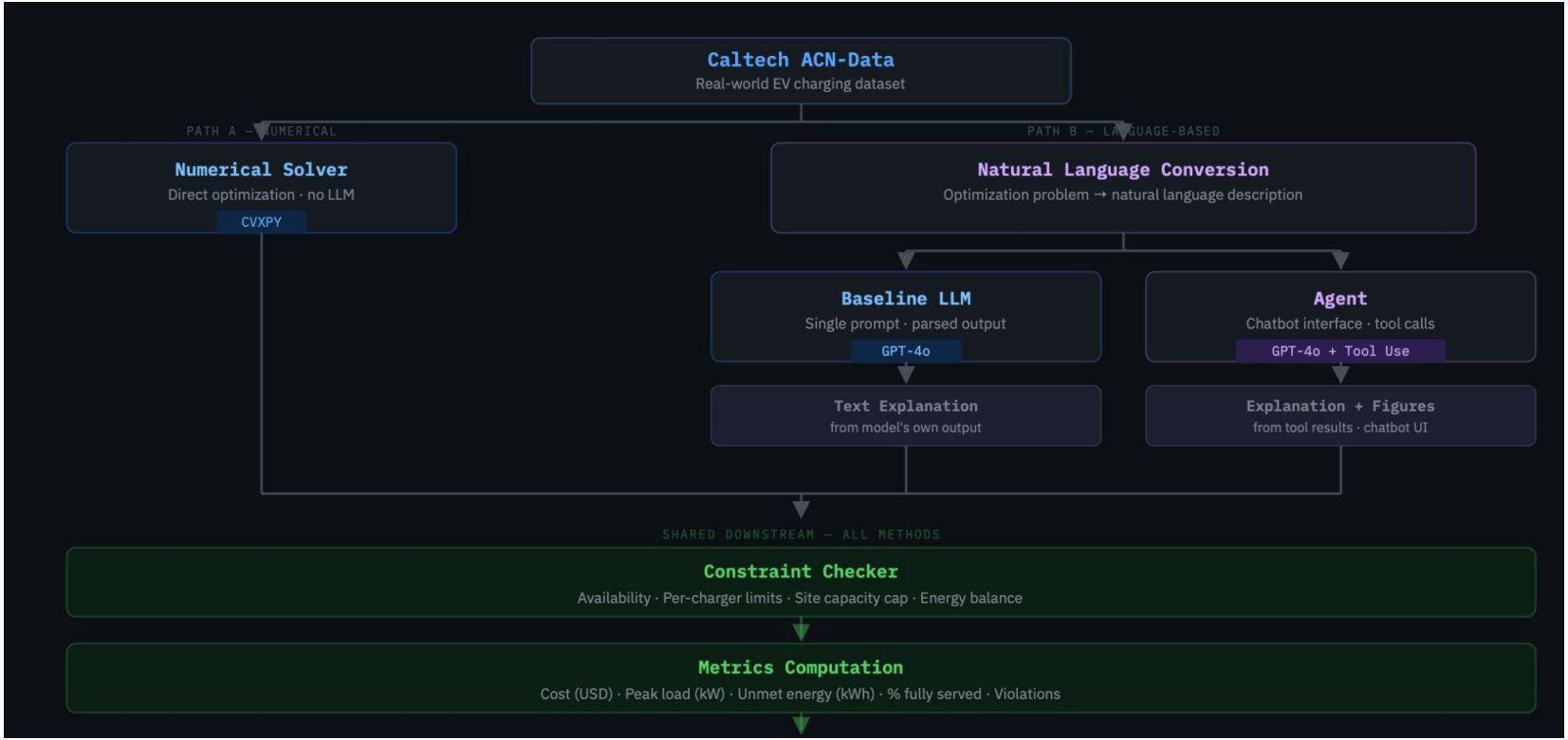


📈 Aggregate Load Profile



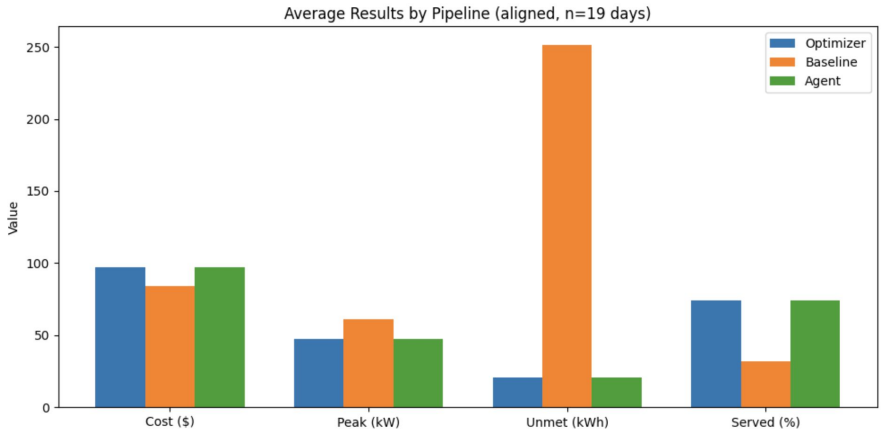
You

Methodology Pipeline



Results

- Agent matches optimizer exactly (same cost, peak load, unmet energy) on 19 benchmark days but adds features
- Agent: 30–100% fully served
- LLM-only: 0–72% fully served (Extremely inconsistent)



LLM Baseline is inconsistent. Some days it performs well, other days it completely fails

```
[4/20] 2019-05-22
    Optimizer: cost=$67.30  served=97.7%
    Agent:     cost=$67.30  served=97.7%
    Baseline:  cost=$192.05  served=58.1%

[5/20] 2019-06-03
    Optimizer: cost=$76.40  served=92.3%
    Agent:     cost=$76.40  served=92.3%
    Baseline:  cost=$189.69  served=71.8%
```

LLM Baseline Serving Well

```
[1/20] 2019-05-01
    Optimizer: cost=$111.57  served=97.7%
    Agent:     cost=$111.57  served=97.7%
    Baseline:  cost=$8.88  served=13.6%

[2/20] 2019-05-08
    Optimizer: cost=$68.99  served=97.9%
    Agent:     cost=$68.99  served=97.9%
    Baseline:  cost=$2.02  served=0.0%
```

LLM Baseline Failing

LLM's frequently under serves to minimize cost thus skyrocketing unmet kWh across 19 days

Pipeline	Days	Avg Cost ($)	Avg Peak (kW)	Avg Unmet (kWh)	Avg Served (%)	Avg Violations
Optimizer	19	97.19	47.21	20.95	73.86	13.11
Baseline	19	84.34	60.97	251.59	31.83	86.11
Agent	19	97.19	47.21	20.95	73.86	13.11

Live Demo

Live Demo

Web UI at `localhost:8000`; natural-language request → agent runs optimization → returns explanation, schedule figures and metrics

Additional Parameter Control:

Agent will consider additional constraints such as

- “provide schedule if max peak load is 5kW”
- “What if EV2 needs an additional 10kWh”

If unable to develop feasible schedule → Generates an explanation of Failure

Visualization

Provides a power schedule by session over time & per EV over time.

Conclusion / Future work

Conclusions

- Agentic LLM + optimizer matches the numerical solver and produces faithful explanations
- Agent accepts NL input whereas LLM baseline requires structured table input and may still fail to produce valid schedule
- LLM-only fails constraint satisfaction

Limitations

- Single-site, day ahead planning setup
- Performance degrades severely if session data or TOU rates are not available

Future work

- Multi-site aggregation
- Peak penalty objective sweep
- Context aware follow up prompts
- Intra-day planning (Closer to live)

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- Brown et al., “Language Models are Few-Shot Learners,” NeurIPS 2020
- Wang et al., “Smart Charging for Electric Vehicles,” IEEE Comm. Surveys 2016
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- Caltech ACN-Data: ev.caltech.edu
- Figure 1:Green Convergence. (n.d.). *What is Time-of-Use?* Retrieved March 9, 2026, from https://www.greenconvergence.com/blog/what-is-time-of-use

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Thank you!